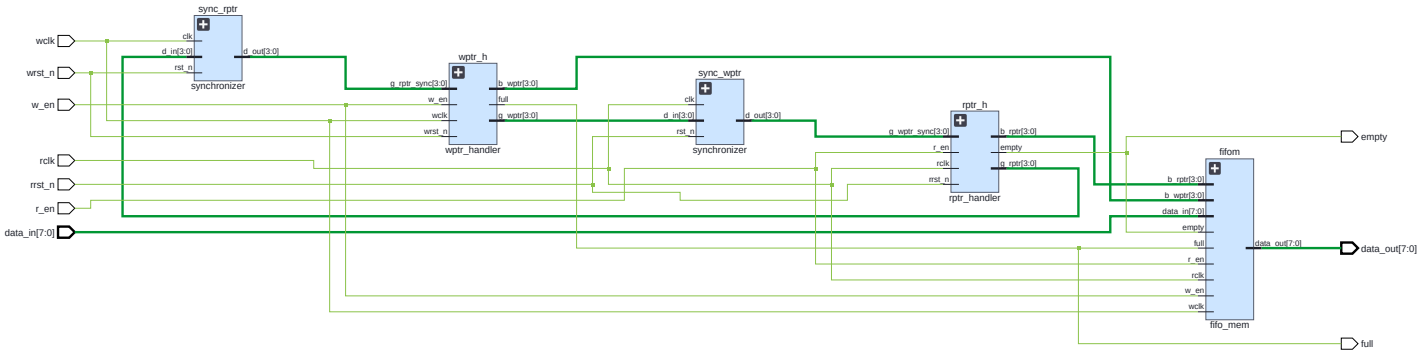
wclk
wrst_n
w_en
rclk
rrst_n
r_en
data_in[7:0]
sync_rptr
clk
d_in[3:0]
rst_n
d_out[3:0]
synchronizer
wptr_h
g_rptr_sync[3:0]
w_en
wclk
wrst_n
b_wptr[3:0]
full
g_wptr[3:0]
wptr_handler
sync_wptr
clk
d_in[3:0]
rst_n
d_out[3:0]
synchronizer
rptr_h
g_wptr_sync[3:0]
r_en
rclk
rrst_n
b_rptr[3:0]
empty
g_rptr[3:0]
rptr_handler
fifom
b_rptr[3:0]
b_wptr[3:0]
data_in[7:0]
empty
full
r_en
rclk
w_en
wclk
data_out[7:0]
fifo_mem
empty
data_out[7:0]
full